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1  -- =====
2  -- Task 4: SQL for Data Analysis
3  -- Ecommerce SQL Database
4  -- =====
5
6  -- 1. CREATE TABLES
7  CREATE TABLE Customers (
8      CustomerID INT PRIMARY KEY,
9      CustomerName VARCHAR(50) NOT NULL,
10     City VARCHAR(50)
11 );
12
13 CREATE TABLE Products (
14     ProductID INT PRIMARY KEY,
15     ProductName VARCHAR(50) NOT NULL,
16     Category VARCHAR(50),
17     Price DECIMAL(10,2) NOT NULL
18 );
19
20 CREATE TABLE Orders (
21     OrderID INT PRIMARY KEY,
22     CustomerID INT,
23     ProductID INT,
24     Quantity INT,
25     OrderDate DATE,
26     FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID),
27     FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
28 );
29
30 -- 2. BASIC SELECT
31 SELECT * FROM Customers;
32
33 -- 3. WHERE CLAUSE
34 SELECT * FROM Products
35 WHERE Price > 3000;
36
37 -- 4. ORDER BY
38 SELECT * FROM Products
39 ORDER BY Price DESC;
40
41 -- 5. GROUP BY with Aggregate Function
42 SELECT ProductID,
43        SUM(Quantity) AS TotalSales
44 FROM Orders
45 GROUP BY ProductID
46 ORDER BY TotalSales DESC;
47
48 -- 6. INNER JOIN
49 SELECT c.CustomerName, p.ProductName, o.Quantity, o.OrderDate
50 FROM Orders o
51 INNER JOIN Customers c ON o.CustomerID = c.CustomerID
52 INNER JOIN Products p ON o.ProductID = p.ProductID
53 ORDER BY o.OrderDate;
54
55 -- 7. LEFT JOIN
56 SELECT c.CustomerName, o.OrderID, o.OrderDate
57 FROM Customers c
58 LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
59 ORDER BY c.CustomerID;
60
61 -- 8. Aggregate Functions
62 SELECT
63     SUM(Quantity) AS Total_Items,
64     AVG(Quantity) AS Avg_Items,
65     COUNT(*) AS Total_Orders
66 FROM Orders;
67
68 -- 9. SUBQUERY
69 SELECT ProductName, Price
70 FROM Products
71 WHERE Price > (SELECT AVG(Price) FROM Products);
72
73 -- 10. VIEW
74 CREATE VIEW Sales_View AS
75 SELECT c.CustomerName, p.ProductName, o.Quantity, o.OrderDate
76 FROM Orders o
77 JOIN Customers c ON o.CustomerID = c.CustomerID
78 JOIN Products p ON o.ProductID = p.ProductID;
79 -- Query the View
80 SELECT * FROM Sales_View ORDER BY OrderDate;
81
82 -- 11. INDEX
83 CREATE INDEX idx_customer ON Orders(CustomerID);

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customers.png

DB Browser for SQLite - Ecommerce_SQL_Database.db

File Edit View Tools Help

New Database

Open Database

Write Changes

Revert Changes

Open Project

Save Project

Attach Database

Close Database

Database StructureBrowse DataEdit PragmasExecute SQL

1

SELECT * FROM Customers;

Results

Messages

	CustomerID	CustomerName	City
	Filter	Filter	Filter
1	1	Amit	Pune
2	2	Priya	Mumbai
3	3	Rahul	Nagpur
4	4	Sneha	Delhi
5	5	Karan	Pune

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